President's Monthly
Research and Development Highlights
"Creating Tomorrow"
MARCH 2017 • VOLUME 12, ISSUE 3

Awards

Sarah Clayton (Digital Scholarship Specialist, OU Libraries) was named a "Super-Synergist" and 2017 "Mover and Shaker" by *Library Journal*. Sarah won in the "Digital Developer" class for her work on the Presidential Dream Course, "Making Modern America: Discovering the Great Depression and New Deal" with David Wrobel (Professor, History) and Keith Gaddie (Chair, Political Science) and for being one of OU Libraries' first Software Carpentry certified instructors. She is one of only 52 people selected from the 350 nominations made from across the country in 2017, and she joins Matt Cook as the second "Mover and Shaker" from the University Libraries in the past two consecutive years. Read her feature at http://tinyurl.com/l5gsdm6.

OU Engineering Professor Receives National Science Foundation Early CAREER Award

A University of Oklahoma Gallogly College of Engineering professor, Steven P. Crossley, is the recipient of a five-year, National Science Foundation Early CAREER Award in the amount of $548,829 for research that can be used to understand catalysts that are important for a broad range of chemical reactions ranging from the production of renewable fuels and chemicals for natural gas processing. The research will be integrated with educational and outreach programs intended for American Indian students, emphasizing the importance of sustainable energy.

"The NSF CAREER award is partly in recognition of the important work that Steve has already done in the field of catalysis. It is one of the highest honors a young faculty member can receive. We look forward to him doing great things in the future," said Brian P. Grady, director of the OU School of Chemical, Biological and Materials Engineering.

Crossley, an assistant professor in the OU School of Chemical, Biological and Materials Engineering, is also a faculty mentor for the American Indian Science and Engineering Society. The project entitled, "SusChEM:CAREER:Using unique synthesis techniques and reaction kinetics to quantify and manipulate catalytically active sites in metal-reducible oxide systems," will provide a detailed understanding of active sites and atom transfer processes involved in catalytic conversion of bio-oil molecules derived from biomass.

"We are proposing a new method to quantify the role of different catalytically active sites under harsh reaction conditions that are commonly challenging to decouple. Our findings should help to clarify confusion in the literature while providing valuable information necessary for improved catalyst design," said Crossley.

Biomass conversion processes typically create a broad range of oxygenated intermediates that are treated further by catalytic processes to remove excess oxygen and build longer chain hydrocarbons attractive as fuel components and chemical intermediates. The efficient conversion requires multifunctional catalysts—typically composed of metal and metal oxide active sites—capable of several simultaneous or sequential reaction steps. While it is well understood that different types of active sites are required for different reactions, the exact nature of those sites and their ideal proximity is not known.

This study will examine those factors by decoupling metal sites from reducible metal oxide sites using carbon nanotube bridges as hydrogen shuttles. By eliminating direct contact between the metal and metal oxide components, and by varying the metal-metal oxide spacing along the carbon nanotubes, the study will provide an opportunity to examine independently two important aspects of bifunctional catalysis on reducible metal oxides: metal-support interactions and hydrogen spillover effects vary with different types of molecules common to biomass deconstruction processes. For more information on the study, contact Crossley at stevencrossley@ou.edu.

President's Monthly
Research and Development Highlights

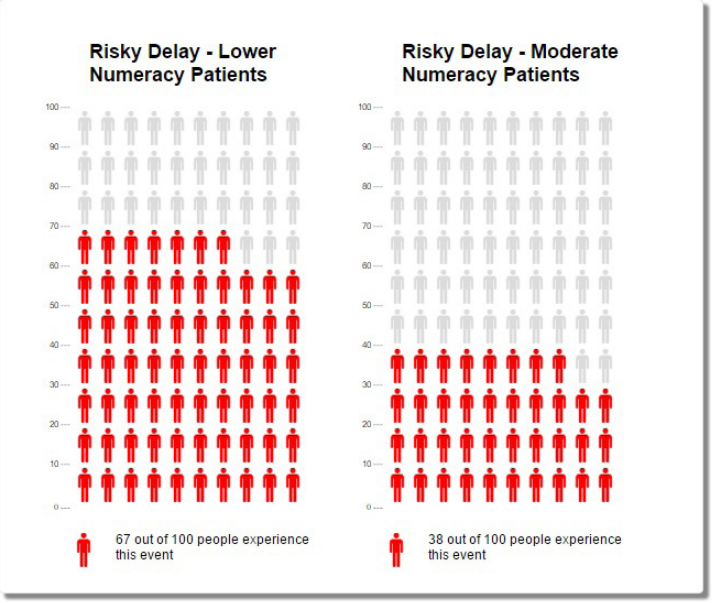
"Creating Tomorrow"

MARCH 2017 • VOLUME 12, ISSUE 3

OU Study Clarifies Risky Decision Making During A Heart Attack

In a recent study to determine why some individuals who experience symptoms for acute coronary syndrome decide to seek medical attention more quickly than others, a University of Oklahoma researcher has identified numeracy—the ability to understand and apply numerical concepts as the primary decision delay risk factor for individuals experiencing the medical condition. Cardiovascular disease, which includes conditions such as acute coronary syndrome, is the number one killer worldwide responsible for about one in three deaths.

Edward Cokely, Presidential Research Professor and associate professor of psychology, OU College of Arts and Sciences and National Institute for Risk & Resilience, and his collaborators followed up with 102 survivors within five days after having experienced ACS. A questionnaire was administered to measure numeracy, decision delay and other relevant factors, such as anxiety, depression, symptom severity, knowledge and demographics. Cokely and collaborators learned that numeracy assessments may predict which patients are at greater risk for life-threating decision delay, which could support development of risk communications for a diverse range of patients who vary in risk literacy.



"We have better insights as to why people do not go to the hospitals, so now we have to find a way to empower these people. Asking specific questions and creating individualized plans or interventions could reduce a person's risk in the future," said Cokely.

Dafina Petrova and Rocio Garcia-Retamero, researchers from the Mind, Brain and Behavior Research Center at the University of Granada, coordinated the international patient study with collaborators from the Cardiology Department at the University Hospital Virgen de las Nieves in Spain. Eligible patients who agreed to participate in the study were identified by a qualified practicing cardiologist who also extracted information regarding final diagnosis and angiogram results. Special care was taken to minimize any burden on patients due to fatigue, illness or other difficulties.

In this study, low patient numeracy was related to longer decision delay, which, in turn, related to higher odds of positive troponin (a protein that indicates heart damage) on arrival at the hospital. A patient with high versus low numeracy was about four times more likely to seek medical attention within the critical first hour after experiencing symptoms. ACS survival rates and outcomes can be improved by up to 50 percent when treatment is administered within one hour of initial symptoms.

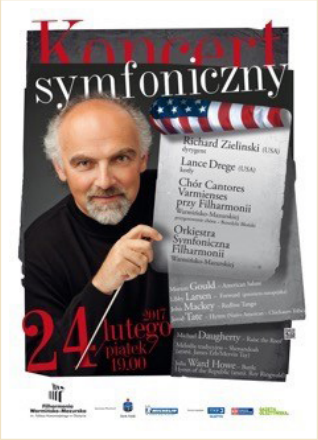
During ACS, the decision to seek medical attention may involve many skills, such as estimating severity and intensity and identifying and evaluating sources of risk precisely and confidently. Acknowledging symptoms as urgent or serious and association of symptoms to a heart attack are two factors that lead to shorter decision delay times. As numeracy fosters an understanding of the ACS risks and tradeoffs, more numerate people are likely to recognize the potential of and need for immediate action.

"Numeracy Predicts Risk of Pre-Hospital Decision Delay: A Retrospective Study of Acute Coronary Syndrome Survival," authored by Cokely and a team of researchers, was published in the Society of Behavioral Medicine's Annals of Behavioral Medicine. Support for this project was provided by the Ministerio de Economia y Competitividad in Spain and the National Science Foundation. For more information about this research project, please contact cokely@ou.edu.

President's Monthly
Research and Development Highlights
"Creating Tomorrow"
MARCH 2017 • VOLUME 12, ISSUE 3

Musical Invitation

By invitations from the Institute of Music at *University of Warmia and Mazury* and the *Warmińsko-Mazurska Filharmonia Orchestra* in Ołsztyn, Poland, School of Music faculty Dr. Richard Zielinski and Dr. Lance Drege presented master classes, seminars and a final concert on February 25, 2017, conducted by Zielinski, highlighting the music of American composers. The final concert featured Dr. Lance Drege, performing the Poland premiere of Michael Daugherty's timpani concerto, RAISE THE ROOF. In addition, the final concert with the Warmińsko-Mazurska Filharmonia Orchestra, *Chór Cantores Varmienses* (a 40-voice professional chorus) presented the European premiere of HYMN, by Oklahoma Chickasaw composer Jerod Impichchaachaaha' Tate.



Governor Mary Fallin Appoints Dr. Kelvin Droegemeier as Secretary of Science and Technology

Governor Mary Fallin announced that Dr. Kelvin Droegemeier, vice president for research at the University of Oklahoma (OU), will be joining her Cabinet as secretary of science and technology.

Droegemeier is a Regents' professor of meteorology, Weathernews chair emeritus, and Teigen presidential professor at OU. He currently serves on Fallin's Science and Technology Council, chairing the academic research and development subcommittee.

Fallin said Droegemeier's extensive background in science and technology, as well as in government, as a key reason for his selection.

"Dr. Droegemeier is a leading expert in the science and engineering research fields, as well as in the education community, and I am thrilled he has agreed to continue to serve the state in a greater capacity," said Fallin. "To realize all that the state and our citizens can become, we must harness the potential of science and technology. This is a part of my focus on 'future ready' education and training, and I know Kelvin will be committed to ensuring Oklahoma makes strides in these critically important areas."

"He is widely respected as a leader on both the local and national levels, especially in encouraging relationships among industry, academia, and government. I have asked him to utilize these skills and his broad depth of knowledge to create new partnerships to benefit all of Oklahoma."

Droegemeier co-founded the National Science Foundation (NSF) Science and Technology Center (STC) for Analysis and Prediction of Storms (CAPS), which is today recognized around the world as the pioneer of storm-scale numerical weather prediction. He also co-founded the NSF Engineering Research Center for Collaborative Adaptive Sensing of the Atmosphere (CASA), and served on the National Science Board under both President George W. Bush and President Barack Obama. He is a Fellow of the American Meteorological Society and the American Association for the Advancement of Science, and, under then-Governor Brad Henry, he chaired the Weather and Climate Team for the EDGE (Economic Development Generating Excellence) Program.

"I am deeply honored by the privilege of serving in Governor Fallin's Cabinet as secretary of science and technology," said Droegemeier. "I look forward to working with the governor, with other Cabinet members, and with elected leaders and citizens in Oklahoma to unlock the full potential of our state's science and technology enterprise. Now is the time for us to seize the moment – to utilize our science and technology assets to transform our economy, create high-paying jobs in future industries, improve the quality of life for all our citizens, and enhance educational opportunities for not only traditional students, but for life-long learners as well."

President's Monthly Research and Development Highlights

"Creating Tomorrow"

MARCH 2017 • VOLUME 12, ISSUE 3

Student Recognition

- Two students in the Cooperative Institute for Mesoscale Meteorological Studies (CIMMS) won awards in the poster presentation category at the American Meteorological Society's 7th Transition of Research to Operations Conference student competition. Katie Wilson (Ph.D. Student) won first place and her poster presentation was titled, "An Instantaneous Self-Assessment of Forecaster Cognitive Workload." Makenzie Krocak (Graduate Student) won second place and her poster presentation was titled, "Establishing a Baseline: What We Know about Tornado Warning Reception, Comprehension and Response."
- Melanie Lewis (Doctoral student, Psychology) won the student division of the third annual ESPN Hackathon in March with her innovative idea to apply an industrial organizational psychology theory that looks at organizational citizenship behavior to sports analytics. Of the 47 students and professionals giving a 90-second presentation in the first round of the hackathon, only three students, including Lewis, and three professionals were selected to give a five-minute presentation at the MIT Sloan Sports Analytics Conference in Boston, Massachusetts.
- Aaron Fournier (J.D. Candidate, Law and Chickasaw Nation Employee, South Central Climate Science Center) was accepted into the Udall Foundation's competitive Native American Congressional Internship Program. Aaron will join 11 other American Indian and Alaska Native Students in Washington D.C. for nine weeks this summer to get an inside look at the government-to-government relationship between Tribes and the federal government and to work alongside American Indian professionals who share his commitment to advocating for tribal sovereignty.
- Elizabeth Smith (Graduate Student, CIMMS) placed first for her oral presentation at the 2017 American Meteorological Society Annual Meeting. Her presentation was "The Great Plains Low-Level Jet During PECAN: Initial Comparisons of Profiling Observations with WRF Model Predictions."

- Several Construction Science (CNS) and Architecture students placed in different categories at the Associated Schools of Construction Region V Student Competition. CNS students coached by Matthew Reyes placed first in the Commercial Building Category. Architecture and CNS students coached by Tammy McCuen placed second in the Design-Build category. Two teams of Architecture and CNS students with Dublin (coached by Somik Ghosh) and Deakin (coached by Anthony Perrenoud) placed second and third, respectively in the Design-Build International category. Chris Parker (Senior, CNS) and Grant Baggs (Senior, CNS) placed first and third respectively as Best Presenters in the Commercial Building category, and Brandon Moews (Junior, CNS) placed third as Best Presenter in the Design-Build International category.

Publication Awards

The article "Contingent Valuation and the Policymaking Process: An Application to Used Nuclear Fuel in the United States" was recently selected as best of articles published in the *Journal of Benefit-Cost Analysis* in 2016. It is authored by political science faculty and researchers, Deven Carlson, Joseph Ripberger, Hank Jenkins-Smith, Carol Silva, and Kuhika Gupta.

President's Monthly Research and Development Highlights

"Creating Tomorrow"

MARCH 2017 • VOLUME 12, ISSUE 3

New Publications

- Cambridge University Press published *Atmospheric Radar: Application and Science of MST Radars in the Earth's Mesosphere, Stratosphere, Troposphere, and Weakly Ionized Regions* by W. Hocking, J. Röttger, Robert Palmer (Executive Director, Advanced Radar Research Center and Professor, Meteorology and Electrical & Computer Engineering), T. Sato, and Phillip Chilson (Professor, Meteorology) (January 2017).
- Oxford University Press published *Organizational Communication* by Michael W. Kramer (Professor & Chair, Communication) and R.S. Bisel (2017).
- Springer published *And the Rest is Just Algebra (1st ed. 2017 Edition)* by Sepideh Stewart (Assistant Professor, Mathematics) (October 2016).

Faculty Recognition

- Ziho Kang (Assistant Professor, Industrial and Systems Engineering) and Steven Landry received the *Andrew P. Sage Best Transaction Paper Award* from the Institute of Electrical and Electronics Engineers (IEEE) Systems, Man, and Cybernetics Society.
- Carsten Schapkow (Associate Professor, History/Schusterman Center for Judaic and Israel Studies) recently became the series editor for *Lexington Studies in Jewish History, Historiography, and Memory.*
- Michael Givel (Professor, Political Science) has been named Director in Bhutan of the Bhutan Rare and Historical Digital Archive Collection by the Kingdom of Bhutan.
- Susan Laird (Professor, Educational Leadership & Policy Studies) completed her work as conference program chair in Seattle, featuring the theme "Love, Labor, and Learning Under the Gun: Education Writ Large with Visionary Pragmatism," in November 2016. She began her one-year term as president of the American Educational Studies Association, a society primarily comprised of college and university professors who teach and research in the field of education utilizing one or more of the liberal arts disciplines of philosophy, history, politics, sociology, anthropology, or economics as well as comparative/international and cultural studies.
- Robert Nairn (Professor, Civil Engineering and Environmental Science) and his team in the Center for Restoration of Ecosystems and Watersheds (CREW) completed construction and began treating metals-contaminated water at the Southeast Commerce Passive Treatment System at the Tar Creek Superfund Site on February 25, 2017. This second OU-designed, off-grid, ecologically-engineered system treats about 100 gallons per minute of iron, zinc, lead, cadmium and arsenic contaminated mine water and uses a 3-kW solar power system for re-aeration and hydrogen sulfide control. Ongoing research will examine the efficiency of specific contaminant removal mechanisms and decreases in ecotoxicity.

- Hazem Refai (Professor, Engineering at OU-Tulsa) and graduate students have been working with ODOT to develop a wireless vehicle detection sensor that will allow ODOT to have the most accurate traffic counts and classifications of cars. This sensor will be used to replace outdated ODOT equipment and will save the state of Oklahoma time, money, and resources while improving traffic conditions across the state. Currently in the fabrication and testing phase, the sensors will provide ODOT with data regarding collision avoidance, vehicle classification, road expansion needs, and volume counts.

President's Monthly
Research and Development Highlights

"Creating Tomorrow"

MARCH 2017 • VOLUME 12, ISSUE 3

February New Awards

Name	Dept./Center	Funding Agency	Project Title	Amount
Mrinal Saha & Yingtao Liu	Aerospace & Mechanical Engineering	Honeywell Federal Manufacturing & Technologies, LLC - Kansas City Plant	Stress Sensing Silicone Films	$105,000
Stephanie Schnorr, Cecil Lewis, & Christina Warinner	Anthropology	L.S.B. Leakey Foundation	Physiological relevance of salivary amylase copy number variation for oral digestion	$20,601
Caryn Vaughn	Biology/OK Biological Survey	National Science Foundation	Collaborative Research: Anchoring adaptations, hydraulic entrainment thresholds, and temporal persistence of imperiled freshwater mussels in river channels with labile beds	$36,645
Lei Ding & Han Yuan	Biomedical Engineering	Hearts for Hearing	Neurophyisological Assessment of Thresholds of Audibility and Loudness in Healthy Persons and Cochlear Implants Users	$100,717
Ming Xue & Guifu Zhang	Center for Analysis & Prediction of Storms/ Meteorology	Beijing Meteorological Service	Assessment of the Performance of Beijing Meteorological Service (BMS) X-band Polarimetric Radars and Data Quality Control and Assimilation for the BMS X-band Radar Network	$120,000
Adam Duerfeldt	Chemistry & Biochemistry	OU Health Sciences Center	Mentoring Diabetes Research in Oklahoma	$50,000
Philip Harvey	Civil Engineering & Environmental Science	WorkSafe Technologies	Shake table testing of WorkSafe Technology ISO-Base and OCTO-Base	$5,000
Justin Reedy	Communication	South Carolina Sea Grant Consortium	Can Deliberative Discussions Lay a Foundation for Integrated Decision Making Networks?	$19,207
Kimberly Elmore	Cooperative Institute for Mesoscale Meteorological Studies	Purdue University	Development of an Automated Winter Road Condition Analysis and Prediction System	$40,593
Henry Neeman	Data Center and Supercomputing	University of Illinois	XSEDE 2.0: Integrating, Enabling and Enhancing National Cyberinfrastructure with Expanding Community Involvement	$67,566
Tian You Yu, Boon Leng Cheong, & Robert Palmer	Electrical & Computer Engineering/Advanced Radar Research Center/ Vice President's Office	National Central University	Technical Support for the Procurement of an S-band Polarimetric Weather Radar	$88,308
Richard Elmore	Geology & Geophysics	Devon Energy Corporation	Diagenetic study of Meramec, Major and Garvin counties, Oklahoma	$97,093

President's Monthly
Research and Development Highlights
"Creating Tomorrow"
MARCH 2017 • VOLUME 12, ISSUE 3

Name	Dept./Center	Funding Agency	Project Title	Amount
Phillip Chilson	Meteorology	Bat Conservation International	Monitoring the Activity of Brazilian Free-Tailed Bats using Weather Radar	$5,000
Xuguang Wang	Meteorology	U.S. Dept. of Defense, Office of Naval Research	A further study of the impact of TCI field campaign observations on hurricane intensification through ensemble-based data assimilation	$135,476
Elizabeth Bergey	OK Biological Survey	Southwestern Oklahoma State University	A Survey for Rare Mayfly and Caddisfly Species of Greatest Conservation Need	$17,443
Xinyu Dai	Physics & Astronomy	Smithsonian Astrophysical Observatory	An Archival Study of Dust, Gas, and Metallicity of High Redshift Galaxies	$48,000
Xinyu Dai	Physics & Astronomy	Smithsonian Astrophysical Observatory	Measuring the Innermost Stable Orbit using Microlensing of the FE K Line	$50,000
Phillip Gutierrez	Physics & Astronomy	Argonne National Laboratory	ANL ASC Graduate Fellowship for OU Graduate Student - Dylan Frizzell	$20,000
Xinyu Dai	Physics & Astronomy	NASA—Headquarters	The Swfit AGN and Cluster Survey	$61,378
John Wisniewski	Physics & Astronomy	NASA—Headquarters	Exploring the Final Stages of Gas Giant Formation	$220,498
Hank Jenkins Smith & Carol Silva	Political Science	Sandia Laboratories	Non-Use Value of Hydropower Publication and Dissemination Project	$60,000
Hollie Mackey	Women's and Gender Studies	Indiana University and Purdue University at Indianapolis	Midwest and Plains Equity Assistance Center: Civil Rights Training and Advisory Services-Equity Assistance Centers Program	$25,000

Number of new awards for the month: **22**
Dollar amount of proposals funded: **$1,393,525**

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